MATH3900 September 15, 2026
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Assignment 1
Due: 11:59 p.m., September 28, 2026

Instructions: *100 pts*

1. Give precise and adequate explanations to support your conclusions. Clarity of presentation counts. You may cite results proved in class without reproving them.
2. Discussions are encouraged. Plagiarism is not allowed. Prepare your own work.
3. Submit one PDF on Gradescope. LATE SUBMISSIONS WILL NOT BE ACCEPTED.
4. Append the AI-use declaration specified at the end, including if you used no AI. **Omitting it results in a deduction of 50 points from the homework mark.**

Throughout, $\|\cdot\|$ denotes the Euclidean norm unless a subscript specifies otherwise.

1. For each statement, determine whether it is true or false. If true, provide a proof; if false, provide a counterexample and verify it.
   (a) Every continuously differentiable function $f : \mathbb{R} \to \mathbb{R}$ that is bounded below has a global minimizer. [4]
   (b) If a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ satisfies $\nabla f(u) = \nabla f(v)$ at two distinct points u and v, then f cannot be strongly convex. [4]
   (c) If a twice continuously differentiable convex function $f : \mathbb{R} \to \mathbb{R}$ has a unique global minimizer, then it is μ-strongly convex for some $\mu > 0$. [4]
   (d) If a convex function $f : \mathbb{R}^d \to \mathbb{R}$ has two distinct global minimizers u and v, then every point on the line segment joining them is a global minimizer. [4]
   (e) If $f : \mathbb{R}^d \to \mathbb{R}$ is continuously differentiable and $\nabla f(x) \neq 0$, then $f(x - \nabla f(x)) < f(x)$. [4]
2. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by
$$f(x_1, x_2) = \frac{1}{4}x_1^4 - \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2.$$
   (a) Compute $\nabla f(x)$ and $\nabla^2 f(x)$, and find all stationary points. [6]
   (b) Classify each stationary point as a local minimizer, a local maximizer, or neither. Find all global minimizers and the minimum value. [8]
   (c) Now let $h(x_1, x_2) = x_1^4 - x_2^4$. Compute its gradient and Hessian at the origin. Is the origin a local minimizer, a local maximizer, or neither? [6]

3. Given observations $a_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$, $i = 1, \ldots, n$, consider the following extension of *ridge regression*:

$$\min_{x \in \mathbb{R}^d} \; f_\lambda(x) := \frac{1}{2n} \sum_{i=1}^{n} (a_i^T x - y_i)^2 + \frac{\lambda}{2} \|x\|^2, \qquad \lambda \in \mathbb{R}.$$

In this question, take $n = d = 2$ with observations

$$a_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad y_1 = 1, \qquad a_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \quad y_2 = 2.$$

(a) For a given $\lambda \in \mathbb{R}$, compute $f_\lambda(x)$ explicitly in terms of x_1 and x_2, and write down its gradient and Hessian. [6]

(b) Determine for which $\lambda \in \mathbb{R}$ the function is convex and for which it is strongly convex. If the function is strongly convex, give a strong-convexity constant $\mu > 0$ and a gradient Lipschitz constant $L > 0$, and compute the ratio $\kappa = L/\mu$ for your choices. [6]

(c) For each $\lambda \in \mathbb{R}$, determine whether a global minimizer exists. If so, find all global minimizers and determine whether the minimizer is unique. For $\lambda = 1$, also compute the minimum value. [8]

4. Consider the objective $f(t) = t^4/4 + t^2/2$ on $\mathbb{R}$ and the gradient descent iteration $t_{k+1} = t_k - \alpha f'(t_k)$, with constant step size $\alpha > 0$.

(a) Compute $f'(t)$ and $f''(t)$, and find all global minimizers. Determine whether f is strongly convex on $\mathbb{R}$ and, if so, give a strong-convexity constant $\mu > 0$. Does its gradient have a finite Lipschitz constant on all of $\mathbb{R}$? Justify your answers. [6]

(b) Starting from $t_0 = 1$, compute t_1 and t_2 when $\alpha = 1/4$. Then determine all $\alpha > 0$ for which the first update from $t_0 = 1$ satisfies $f(t_1) < f(t_0)$. [8]

(c) Fix $\alpha = 1/4$ and now allow any starting point with $|t_0| \le 1$. Show that $|t_k| \le (3/4)^k |t_0|$ for every integer $k \ge 0$. Deduce that they converge to the minimizer. [6]

5. In this question, we derive sign gradient descent from a steepest descent perspective. For $v \in \mathbb{R}^d$, define $\|v\|_1 = \sum_{i=1}^d |v_i|$ and $\|v\|_\infty = \max_i |v_i|$. Define $\operatorname{sign}(t) = 1$ for $t > 0$, $\operatorname{sign}(t) = -1$ for $t < 0$, and $\operatorname{sign}(0) = 0$; apply it to vectors componentwise. Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable.

(a) For an arbitrary $x \in \mathbb{R}^d$, solve $\min_{\|p\|_\infty \le 1} \nabla f(x)^T p$. Find its minimum value and describe all minimizing directions. Determine whether every minimizing direction is a descent direction at x when $\nabla f(x) \neq 0$. [4]

(b) Consider $\phi(t) = t^2/2$ on $\mathbb{R}$ and a fixed step size $\eta > 0$. Starting from $t_0 = \eta/2$, compute t_1 and t_2 for the iteration $t_{k+1} = t_k - \eta \operatorname{sign}(\phi'(t_k))$. Does this sequence converge to the minimizer? Justify your answer. [4]

(c) Assume f is L-smooth with $L > 0$. Find a step-size rule $\alpha = \alpha(\nabla f(x), L, d) \geq 0$ such that, for every $x \in \mathbb{R}^d$, [6]

$$f(x - \alpha \operatorname{sign}(\nabla f(x))) \leq f(x) - \frac{\|\nabla f(x)\|_1^2}{2Ld}.$$

Prove that your rule satisfies this bound. State the corresponding iterative algorithm.

(d) Assume in addition that f is μ-strongly convex, where $0 < \mu \leq L$, and let $x^\star$ be its minimizer. Show that the algorithm in part (c) satisfies, for every integer $k \geq 0$, [6]

$$f(x_k) - f(x^\star) \leq \left(1 - \frac{\mu}{Ld}\right)^k \big(f(x_0) - f(x^\star)\big).$$

AI-use declaration (required). Append a short statement using the highest level that describes your AI use on any part:

0. No AI tools used.
1. Limited: brainstorming, concept explanations, or language editing.
2. Substantial: help developing a proof, calculation, or solution.
3. Extensive: AI generated most of the initial draft of an answer.

Use this format, naming the affected parts and how you checked the assistance:

AI use: Level [0/1/2/3]. Tool(s): [names, or "none"]. Parts and purpose: [brief description, or "none"]. Checks: [how you verified the mathematics, or "not applicable"]. I take responsibility for my submitted work.

A Level 0 declaration is required if you used no AI. **Omitting the declaration results in a deduction of 50 points from the homework mark.**

Level 1, [Cursor (deepseek v4.1 flash)] help me summary the theorems in the slides, generated a cheatsheet. I take responsibility for my submitted work.

P1

(a) F. $f(x) = e^x$, $\forall x\ f(x) > 0$ but no minimizer.

(b) T.

Suppose f is μ-strong convex, $v - u = d$

$$f(v) - f(u) - \nabla f(u)^T (v-u) \geq \frac{\mu}{2}\|v-u\|$$

$$= \int_0^1 \nabla f(u + t(v-u))^T (v-u)\, dt - \nabla f(u)^T (v-u) - \frac{\mu}{2}\|v-u\|$$

$\because f$ is convex and $\nabla f(u) = \nabla f(v)$

Let $g : [0,1] \to \mathbb{R}$. $g(t) = f(u + t(v-u))$. $\forall \theta, x, y \in [0,1]$.

and $T = \theta x + (1-\theta) y$.

$$g(T) = f(u + T(v-u))$$

$$= f[\theta(u + x(v-u)) + (1-\theta)(u + y(v-u))]$$

$$\leq \theta f(u + x(v-u)) + (1-\theta) f(u + y(v-u)) \qquad = \nabla f(u)^T (v-u)$$

$$= \theta g(x) + (1-\theta) g(y)$$

$\therefore g(t)$ is convex.

$\therefore g'(t) = g'(0) = g'(1)$

$\therefore \nabla f(u + t(v-u))^T (v-u) = g'(t) = g'(0) = \nabla f(u)^T (v-u) \quad \forall t \in [0,1]$

$\therefore$ LHS $= \int_0^1 \nabla f(u)^T (v-u)\, dt - \nabla f(v)^T (v-u) - \frac{\mu}{2}\|v-u\|$

$$= 0 - \frac{\mu}{2}\|v-u\| < 0$$

which contradict to μ-strong convex

(c) F. Consider $f(x) = x^4$.

$$f(v) - f(0) - f'(0)(v-0) - \frac{\mu}{2}(v-0)^2$$

$$= v^2(v^2 - \mu).$$

$\forall \mu > 0$, we can find $v = \frac{1}{2}\sqrt{\mu}$, s.t. LHS < 0.

(d) T $\nabla f(u) = \nabla f(v) = 0$.

$\forall t \in [0,1]$. $f(u + t(v-u)) - f(u)$

$$= \int_0^1 \nabla f(u+t(v-u))^T (v-u)\, dt$$

According to P_1 (b), $\nabla f(u+t(v-u))^T (v-u) = \nabla f(u)^T (v-u)$.

$\therefore$ RHS $= \int_0^1 \nabla f(u)^T (v-u)\, dt = 0$.

i.e. $f(u + t(v-u)) = f(u)$ $\forall t \in [0,1]$ is a global minimizer.

(e) F. Consider $f(x) = x^2$. $f(x - \nabla f(x)) = (x - 2x)^2 = x^2 = f(x)$.

P2.

(a) $\nabla f(x) = (x_1^3 - x_1, x_2)$

$$\nabla^2 f(x) = \begin{pmatrix} 3x_1^2 - 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Let $\nabla f(x) = 0$. $\begin{cases} x_1^3 - x_1 = 0 \\ x_2 = 0 \end{cases} \rightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$ or $\begin{cases} x_1 = 1 \\ x_2 = 0 \end{cases}$

i.e. $(0,0)$, $(\pm 1, 0)$ are stationary points.

(b)

① $x=(0,0)$ $\nabla^2 f(x) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ is neither positive def nor negative def.

② $x=(\pm 1,0)$ $\nabla^2 f(x) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ all eigenvalues are positive.

ie. $\nabla^2 f(x) \succ 0$

$\therefore (\pm 1, 0)$ is local minimizer

(c) $h(0,0) = 0$.

$\forall \varepsilon > 0$, we can find $t < \varepsilon$.

$h(\vec{0} + t(0,1)) = -t^4 < h(0,0)$ and $\|t(0,1)\| = t < \varepsilon$

and $h(\vec{0} + t(1,0)) = t^4 > h(0,0)$ and $\|t(0,1)\| = t < \varepsilon$

$\therefore (0,0)$ is neither.

P3

(a) $f_\lambda(x) = \frac{1}{4}(x_1+x_2-1)^2 + \frac{1}{4}(2x_1+2x_2-2)^2 + \frac{\lambda}{2}(x_1^2+x_2^2)$

$= \frac{2\lambda+5}{4}x_1^2 + \frac{2\lambda+5}{4}x_2^2 + \frac{5}{2}x_1x_2 - \frac{5}{2}x_1 - \frac{5}{2}x_2 + \frac{5}{4}$.

$\nabla f_\lambda(x) = \left(\frac{2\lambda+5}{2}x_1 + \frac{5}{2}x_2 - \frac{5}{2},\ \frac{2\lambda+5}{2}x_2 + \frac{5}{2}x_1 - \frac{5}{2}\right)$

$\nabla^2 f_\lambda(x) = \begin{pmatrix} \frac{2\lambda+5}{2} & \frac{5}{2} \\ \frac{5}{2} & \frac{2\lambda+5}{2} \end{pmatrix}$

(b) Let $g(x) = f_\lambda(x) - \frac{\mu}{2}\|x\|^2$

$= f_{\lambda-\mu}(x)$

so let $\lambda' = \lambda - \mu$. then $f_\lambda(x)$ μ-strong convex $\Leftrightarrow$ $g(x)$ convex

$\Leftrightarrow f_{\lambda'}(x)$ convex.

$\nabla^2 f_{\lambda'}(x) = \begin{pmatrix} \frac{2\lambda'+5}{2} & \frac{5}{2} \\ \frac{5}{2} & \frac{2\lambda'+5}{2} \end{pmatrix}$

Let $v = (a, b) \in \mathbb{R}^2$.

$v^T \nabla^2 f_{\lambda'}(x) v = (a\ b)\nabla^2 f_{\lambda'}(x)\begin{pmatrix} a \\ b \end{pmatrix}$

$= a\left(\frac{2\lambda'+5}{2}a + \frac{5}{2}b\right) + b\left(\frac{5}{2}a + \frac{2\lambda'+5}{2}b\right)$

$= \frac{2\lambda'+5}{2}(a^2+b^2) + 5ab$

$= \lambda'(a^2+b^2) + \frac{5}{2}(a+b)^2 \geq 0$ iff $\lambda' \geq 0$

i.e. $f_{\lambda'}(x)$ convex $\leftrightarrow \nabla^2 f_\lambda(x) \succeq 0 \Leftrightarrow \lambda \geq 0$.

so $\lambda' = \lambda - \mu \geq 0$ i.e. $f_\lambda(x)$ is λ-strong convex. when $\lambda > 0$.

$\forall u, v \in \mathbb{R}^2$.

$$\|\nabla f(u) - \nabla f(v)\|^2 = \left\|\left(\frac{2\lambda+5}{2}(u_1-v_1) + \frac{5}{2}(u_2-v_2),\ \frac{2\lambda+5}{2}(u_2-v_2) + \frac{5}{2}(u_1-v_1)\right)\right\|^2$$

$$= \frac{(2\lambda+5)^2+25}{4}\|u-v\|^2 + \frac{5(2\lambda+5)}{2}(u_1-v_1)(u_2-v_2)$$

By Cauchy's inequality, we have

$$RHS \le \frac{(2\lambda+5)^2+25}{4}\|u-v\|^2 + \frac{5(2\lambda+5)}{2}\cdot\frac{(u_1-v_1)^2+(u_2-v_2)^2}{2}$$

$$= \frac{\left((2\lambda+5)+5\right)^2\|u-v\|^2}{4}$$

$$= (\lambda+5)^2\|u-v\|^2.$$

i.e. $\forall u, v \in \mathbb{R}^2$, $\|\nabla f(u) - \nabla f(v)\| \le |\lambda+5|\,\|u-v\| = (\lambda+5)\|u-v\|$

i.e. $f_\lambda(x)$ is $|\lambda+5|$-smooth.

$\therefore K = \frac{L}{\mu} = \frac{\lambda+5}{\lambda}$ when $\lambda > 0$

(c) Let $\nabla f_\lambda(x) = 0$.

$$\begin{cases} \frac{2\lambda+5}{2}x_1 + \frac{5}{2}x_2 - \frac{5}{2} = 0 & \cdots ① \\ \frac{2\lambda+5}{2}x_2 + \frac{5}{2}x_1 - \frac{5}{2} = 0 & \cdots ② \end{cases}$$

① − ②: $\lambda(x_1 - x_2) = 0$ ⋯ ③

① + ②: $(\lambda+5)(x_1+x_2) = 5$ ⋯ ④

$\because \lambda > 0 \quad \therefore x_1 = x_2$ by ③.

then substitude into ④, we get: $x_1 = x_2 = \frac{5}{2\lambda+10}$ is unique.

$\therefore \left(\frac{5}{2\lambda+10}, \frac{5}{2\lambda+10}\right)$ is the global minimizer (unique) because $f_\lambda(x)$ is convex.

For $\lambda = 1$, $\min_x f_1(x) = f_1\left(\frac{5}{12}, \frac{5}{12}\right) = \frac{5}{24}$

P4 (a) $f'(t) = t^3 + t$

$f''(t) = 3t^2 + 1 > 0$ i.e. $f(t)$ is convex

Let $f'(t) = 0$ i.e. $t^3 + t = t(t^2+1) = 0$.

The only solution is $t = 0$.

$f''(0) = 1 > 0$ $\therefore$ $f(0) = 0$ is the unique global minimizer

$g(t) = f(t) - \frac{\mu}{2}t^2$.

then $g'(t) = t^3 + (1-\mu)t$.

$g''(t) = 3t^2 + (1-\mu) \geq 0$ iff $\mu \leq 1$.

$\therefore$ $f(t)$ is 1- strong convex. $\mu = 1$

$|f'(u) - f'(v)| = |u^3 + u - v^3 - v|$

$= |(u-v)(u^2+uv+v^2+1)|$

$= |u^2+uv+v^2+1| \cdot |u-v|$

$\forall \varepsilon > 0$, let $u = 0$, $v = \varepsilon + 1$.

then RHS $= \left((\varepsilon+1)^2+1\right)|u-v| > \varepsilon |u-v|$.

$\therefore$ $f(t)$ isn't smooth.

$\frac{1}{4} + \frac{1}{2}^2$

(b) ① $f'(t_0) = |+1| = 2$.

$\therefore t_1 = t_0 - \alpha f'(t_0) = 1 - \frac{2}{4} = \frac{1}{2}$

$f'(t_1) = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}$

$\therefore t_2 = t_1 - \alpha f'(t_1) = \frac{1}{2} - \frac{5}{32} = \frac{11}{32}$

② $f(t_0) = \frac{1^4}{4} + \frac{1^2}{2} = \frac{3}{4}$.

$f(t_1) - f(t_0) = f(t_0 - \alpha f'(t_0)) - f(t_0)$

$= f(1-2\alpha) - f(t_0)$

$= \frac{(1-2\alpha)^4}{4} + \frac{(1-2\alpha)^2}{2} - \frac{3}{4}$

$= \frac{1}{4}\left[(1-2\alpha)^2+3\right]\left[(1-2\alpha)^2-1\right] \leq 0$.

$\therefore -3 \leq (1-2\alpha)^2 \leq 1$

i.e. $0 < \alpha < 1$

$\therefore f(t_1) < f(t_0)$ when $\alpha \in (0,1)$

(c)

Base case : $|t_0| \le 1 \le (\frac{3}{4})^0 \cdot |t_0|$

Induction step : suppose $|t_k| \le (\frac{3}{4})^k |t_0|$. then we have :

$$
\begin{aligned}
&|t_{k+1}| - \tfrac{3}{4}|t_k| \\
&= |t_k - \alpha f'(t_k)| - \tfrac{3}{4}|t_k| \\
&= |t_k - \tfrac{1}{4}(t_k^3 + t_k)| - \tfrac{3}{4}|t_k| \\
&= |-\tfrac{1}{4}t_k^3 + \tfrac{1}{4}t_k| - |\tfrac{3}{4}t_k| \\
&= \frac{|t_k|}{4}\left(|3 - t_k^2| - 3\right)
\end{aligned}
$$

$\because |t_k| \le (\frac{3}{4})^k |t_0| \le (\frac{3}{4})^k$

$\therefore t_k^2 \le (\frac{3}{4})^{2k}$ i.e. $t_k^2 \in [0, (\frac{3}{4})^{2k}]$

i.e. $\frac{|t_k|}{4}\left(|3 - t_k^2| - 3\right) \le 0$, equals when $t_k = t_{k-1} = \cdots = t_0 = 0$

$\therefore |t_{k+1}| \le \frac{3}{4}|t_k| \le (\frac{3}{4})^{k+1}|t_0|$.

$\therefore \exists \varepsilon > 0$, we can find $K = \left\lceil \frac{\ln \varepsilon \ln 4}{\ln 3} \right\rceil + 1$, s.t. $|t_{k+1}| \le (\frac{3}{4})^k |t_0| \le (\frac{3}{4})^k \le \frac{3}{4}\varepsilon < \varepsilon$.

i.e. $\lim_{k \to \infty} t_k = 0$. which is global minimizer of $f(t)$.

P5

(a) Let $g = \nabla f(x)$.

$$g^T p = \sum g_i p_i \geq \sum -|g_i||p_i|$$

$$\geq \sum -|g_i|$$

$$= \sum -g_i \cdot sign(g_i)$$

$$\therefore p = (sign(g_1), \cdots, sign(g_d))$$

and $\min_{\|p\|_\infty \leq 1} \nabla f(x)^T p = -\|\nabla f(x)\|_1$

p' is a minimizing direction if:

$$p_i = \begin{cases} sign(g_i) & , g_i \neq 0 \\ \text{any value in } [0,1] & , g_i = 0 \end{cases}$$

s.t. $\nabla f(x)^T p = -\|\nabla f(x)^T\|_1$ is a minimum value.

(b) $t_0 = \frac{\eta}{2} > 0$

$$t_1 = t_0 - \eta \cdot sign(t_0) = -\frac{\eta}{2}$$

$$t_2 = t_1 - \eta \cdot sign(t_1) = \frac{\eta}{2} = t_0.$$

$$\therefore t_k = \begin{cases} \frac{\eta}{2} & , k \text{ is even} \\ -\frac{\eta}{2} & , k \text{ is odd} \end{cases}$$ never converge.

(c) By the descent lemma,

$$f(x) - f(x - \alpha\,\mathrm{sign}(\nabla f(x))) - \frac{\|\nabla f(x)\|_1^2}{2Ld}$$

$$\geq \nabla f(x)^T(\alpha\,\mathrm{sign}(\nabla f(x))) - \frac{L}{2}\|\alpha\,\mathrm{sign}(\nabla f(x))\|^2 - \frac{\|\nabla f(x)\|_1^2}{2Ld}.$$

$$= \alpha\cdot\|\nabla f(x)\|_1 - \frac{L}{2}\|\alpha\cdot\mathrm{sign}(\nabla f(x))\|^2 - \frac{\|\nabla f(x)\|_1^2}{2Ld}.$$

Let $\alpha = \frac{\|\nabla f(x)\|_1}{Ld}$. then we have:

$$RHS = \frac{1}{Ld}\cdot\|\nabla f(x)\|_1^2 - \frac{L}{2}\cdot\frac{1}{L^2d^2}\|\nabla f(x)\|_1^2 - \frac{\|\nabla f(x)\|_1^2}{2Ld}$$

$$= \|\nabla f(x)\|_1^2\cdot\frac{2d-1-d}{Ld^2} \geq 0.$$

Algorithm: $x_0 \in \mathbb{R}^d$ $\epsilon \geq 0$, $\alpha(L,d,\nabla f(x)) = \frac{\|\nabla f(x)\|_1}{Ld}$

for $k = 0, 1, \cdots$ do

if $\|\nabla f(x_k)\| \leq \epsilon$ then return x_k.

$x_{t+1} = x_t - \frac{1}{Ld}\|\nabla f(x)\|_1\cdot\mathrm{sign}(\nabla f(x))$

end for

(d) We want to prove: $f(x_{k+1}) - f(x^*) \leq (1 - \frac{\mu}{Ld})(f(x_k) - f(x^*))$

i.e. $f(x_k) - f(x_{k+1}) \geq \frac{\mu}{Ld}(f(x_k) - f(x^*))$

According to P5 (c),

LHS > ... ①

By the property of μ-strong convex function:

$$RHS \leq \frac{\mu}{Ld} \cdot \frac{\|\nabla f(x)\|^2}{2\mu}.$$

∴ We just need to prove:

$$\frac{\mu}{Ld} \cdot \frac{\|\nabla f(x)\|^2}{2\mu} \leq \frac{\|\nabla f(x)\|_1^2}{2Ld}$$

i.e. $\|\nabla f(x)\|^2 = \sum |\nabla f(x)|_i^2 \leq \left(\sum |\nabla f(x)_i|\right)^2$.

Q.E.D